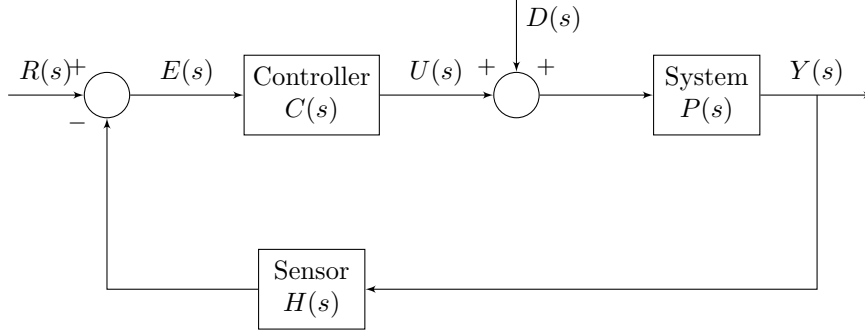


System
Disturbance
Feedback loop
open close System



1 Linear Constant Coefficient Ordinary Differential Equations (LCC-ODE)

A differential equation of the form

$$\sum_{k=0}^n a_k \frac{d^k y(t)}{dt^k} = \sum_{m=0}^M b_m \frac{d^m r(t)}{dt^m},$$

where $y(t)$ is the output (response) of the system and $r(t)$ is the input to the system; and a_i and b_i are constant.

2 Laplace Transform

The Laplace transform of a function $f(t)$, defined for all real numbers $t \geq 0$, is the function $F(s)$, which is a unilateral transform defined by

$$\mathcal{L}[f](s) = F(s) = \int_{0^-}^{\infty} f(t) e^{-st} dt,$$

where $s = \sigma + i\omega$ complex frequency-domain parameter, and, 0^- is the shorthand notation for

$$\lim_{\varepsilon \rightarrow 0^+} \int_{-\varepsilon}^{\infty}.$$

2.1 Region of Convergence (RoC)

2.2 Properties of Laplace Transform

2.3 Standard Laplace Transform Pairs

Function	Time Domain $f(t)$	Laplace Domain $F(s)$	ROC
Impulse	$\delta(t)$	1	All s
Step	$u(t)$	$\frac{1}{s}$	$\Re(s) > 0$
Ramp	$tu(t)$	$\frac{1}{s^2}$	$\Re(s) > 0$
Power	$t^n u(t)$	$\frac{n!}{s^{n+1}}$	$\Re(s) > 0$
Exponential	$e^{-at}u(t)$	$\frac{1}{s+a}$	$\Re(s) > -a$
Sine	$\sin(\omega t)u(t)$	$\frac{\omega}{s^2 + \omega^2}$	$\Re(s) > 0$
Cosine	$\cos(\omega t)u(t)$	$\frac{s}{s^2 + \omega^2}$	$\Re(s) > 0$

3 Second-Order System Analysis

Consider a linear time-invariant (LTI) second-order system described by the transfer function $G(s)$:

$$G(s) = \frac{b}{s^2 + as + b}$$

where a and b are constant coefficients. To analyze the dynamic behavior, we map these coefficients to standard physical parameters.

Definition 3.1 (Natural Frequency, ω_n): The natural frequency, ω_n , represents the frequency at which the system would oscillate in the absence of damping. It is defined as:

$$\omega_n := \sqrt{b}$$

Definition 3.2 (Damping Ratio, ζ): The damping ratio, ζ , is a dimensionless parameter that characterizes the rate at which oscillations decay. It is related to the system coefficients by:

$$\zeta := \frac{a}{2\omega_n} = \frac{a}{2\sqrt{b}}$$

Substituting these parameters back into the original expression yields the **canonical form** of the second-order transfer function:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

The poles of the system, denoted as $s_{1,2}$, are the roots of the characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$. Using the quadratic formula, the poles are given by:

$$s_{1,2} = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

4 Underdamped Second Order System

$$Y(s) = \frac{1}{s} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{A_1}{s} + \frac{A_2 s + A_3}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$Y(s) = \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)} - \frac{(\zeta/\sqrt{1 - \zeta^2})\omega_n\sqrt{1 - \zeta^2}}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)}$$

$$\omega_d := \omega_n\sqrt{1 - \zeta^2}$$

$$y(t) = \left(1 - e^{-\zeta\omega_n t} \cos(\omega_d t) - \frac{\zeta}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t)\right) u(t)$$

Definition 4.1 (Rise time, T_r): The time required for the waveform to go from 0.1 of the final value to 0.9 of the final value.

$$T_r = \frac{\pi - \theta}{\omega_n\sqrt{1 - \zeta^2}}, \quad \theta = \arctan\left(\frac{\sqrt{1 - \zeta^2}}{\zeta}\right)$$

Definition 4.2 (Peak time, T_p): The time required to reach the first, or maximum, peak.

$$T_p = \frac{\pi}{\omega_n\sqrt{1 - \zeta^2}}$$

Definition 4.3 (Max overshoot, OS): The amount that the waveform overshoots the steady-state, or final, value at the peak time, expressed as a ratio of the steady-state value.

$$OS = \exp\left(-\zeta\pi/\sqrt{1 - \zeta^2}\right)$$

Definition 4.4 (Settling time, T_s): The time required for the transient's damped oscillations to reach and stay within $\pm 2\%$ of the steady-state value.

$$T_s = \frac{-\ln\left(0.02\sqrt{1 - \zeta^2}\right)}{\zeta\omega_n} \approx \frac{4}{\zeta\omega_n}$$

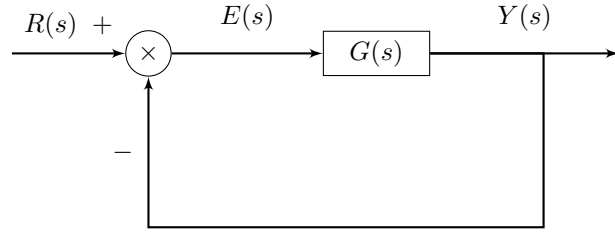
5 Second Order System With Additional Pole

6 Feedback Loop

Definition 6.1 (Feedback Loop Transfer Function):

$$T(s) := \frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

7 Unity Feedback Steady State Error



Consider a unity feedback system with forward transfer function $G(s)$. The closed-loop structure is

$$Y(s) = G(s)E(s), \quad E(s) = R(s) - Y(s).$$

Definition 7.1 (Steady State Error): Suppose the closed-loop system is asymptotically stable and the limits exist. The steady-state error is defined as

$$e_{ss} := \lim_{t \rightarrow \infty} e(t).$$

If $e(t)$ admits a Laplace transform $E(s)$ and the Final Value Theorem applies, then

$$e_{ss} = \lim_{s \rightarrow 0} sE(s).$$

Theorem 7.2 (Steady-State Error for Unity Feedback): For a unity feedback system with forward transfer function $G(s)$, if the closed-loop system is asymptotically stable and the Final Value Theorem applies, then

$$e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)}.$$

Proof. From the unity feedback structure,

$$E(s) = R(s) - Y(s).$$

Since $Y(s) = G(s)E(s)$, we substitute to obtain

$$E(s) = R(s) - G(s)E(s).$$

Rearranging,

$$E(s)(1 + G(s)) = R(s).$$

Hence,

$$E(s) = \frac{R(s)}{1 + G(s)}.$$

Assume:

1. The closed-loop system is asymptotically stable.
2. $sE(s)$ has no poles in the closed right half-plane except possibly at $s = 0$.

Then the Final Value Theorem applies, and

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s).$$

Substituting the expression for $E(s)$,

$$e_{ss} = \lim_{s \rightarrow 0} s \frac{R(s)}{1 + G(s)}.$$

This proves the result. □

7.0.1 Unity Feedback Steady State Error for Step Input

7.0.2 Unity Feedback Steady State Error for Ramp Input

7.0.3 Unity Feedback Steady State Error for Parabola Input

7.1 Non Unity Feedback Steady State Error

7.2 Steady State Error with Disturbance